

安徽大学 2021—2022 学年第二学期

《高等数学 C》期中试题参考答案及评分标准

一、选择题(每小题3分, 共15分)

1.C 2.D 3.B 4.C 5.B

二、填空题(每小题3分, 共15分)

6. $[-1, 2]$ 7. 0 8. $\frac{1}{2} \ln 2$ 9. $-f'(\cos x) \sin x$ 10. -2

三、计算题(每小题10分, 共60分)

$$11. \text{解: } \frac{\frac{n(n+1)}{2}}{2n^2+1} \leq \frac{1}{2n^2+1} + \frac{2}{2n^2+2} + \cdots + \frac{n}{2n^2+n} \leq \frac{\frac{n(n+1)}{2}}{2n^2+n}$$

$$\text{又 } \lim_{n \rightarrow \infty} \frac{\frac{n(n+1)}{2}}{2n^2+1} = \lim_{n \rightarrow \infty} \frac{\frac{n(n+1)}{2}}{2n^2+n} = \frac{1}{4}, \text{ 由夹逼准则, 得}$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2n^2+1} + \frac{2}{2n^2+2} + \cdots + \frac{n}{2n^2+n} \right) = \frac{1}{4} \dots\dots\dots (10 \text{ 分})$$

$$12. \text{解: } \lim_{x \rightarrow \infty} \left(\frac{x+1}{x-1} \right)^x = \lim_{x \rightarrow \infty} \left(1 + \frac{2}{x-1} \right)^{\frac{x-1}{2} \cdot \frac{2}{x-1} x} = e^{\lim_{x \rightarrow \infty} \frac{2x}{x-1}} = e^2 \dots\dots\dots (10 \text{ 分})$$

$$13. \text{解: } \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x \sin^2 x} = \lim_{x \rightarrow 0} \frac{\tan x (1 - \cos x)}{x^3} = \lim_{x \rightarrow 0} \frac{x \cdot \frac{1}{2} x^2}{x^3} = \frac{1}{2} \dots\dots\dots (10 \text{ 分})$$

14. 解: 间断点为 $x = -1, x = 3$

$$\text{由 } \lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 - 2x - 3} = \lim_{x \rightarrow -1} \frac{(x+1)(x-1)}{(x+1)(x-3)} = \lim_{x \rightarrow -1} \frac{x-1}{x-3} = \frac{1}{2}$$

故 $x = -1$ 是第一类可去间断点

$$\text{由 } \lim_{x \rightarrow 3} \frac{x^2 - 1}{x^2 - 2x - 3} = \lim_{x \rightarrow 3} \frac{(x+1)(x-1)}{(x+1)(x-3)} = \lim_{x \rightarrow 3} \frac{x-1}{x-3} = \infty$$

故 $x = 3$ 是第二类无穷间断点..... (10分)

$$\begin{aligned} 15. \text{解: } y' &= \frac{1}{1 + \left(\frac{x+1}{x-1} \right)^2} \left(\frac{x+1}{x-1} \right)' = \frac{1}{1 + \left(\frac{x+1}{x-1} \right)^2} \left(1 + \frac{2}{x-1} \right)' \\ &= \frac{1}{1 + \left(\frac{x+1}{x-1} \right)^2} \cdot \left(-\frac{2}{(x-1)^2} \right) = -\frac{1}{x^2 + 1} \dots\dots\dots (10 \text{ 分}) \end{aligned}$$

16.解：对方程两边求导

$$e^y y' + y + xy' = 0 \Rightarrow y'(x) = -\frac{y}{e^x + x}$$

当 $x = 0$ 时，由原方程得 $y = 1$,

将其代入上式，得 $y'(0) = -\frac{1}{e}$(10分)

四、计算题(每小题10分，共10分)

17.解：由题意知 $y = \frac{2}{x}$,得 $y' = -\frac{2}{x^2}$

设切点为 $(x_0, \frac{2}{x_0})$,则切线方程为 $y - \frac{2}{x_0} = -\frac{2}{x_0^2}(x - x_0)$

当 $x = 0$ 时, $y = \frac{4}{x_0}$;当 $y = 0$ 时, $x = 2x_0$

故 $S = \frac{1}{2} \left| \frac{4}{x_0} \right| |2x_0| = 4$(10分)